

RHAPSODY: A Dataset for Highlight Detection in Podcasts

Younghan Park, Anuj Diwan, David Harwath*, Eunsol Choi*



연세대학교
YONSEI UNIVERSITY



TEXAS
The University of Texas at Austin



NYU

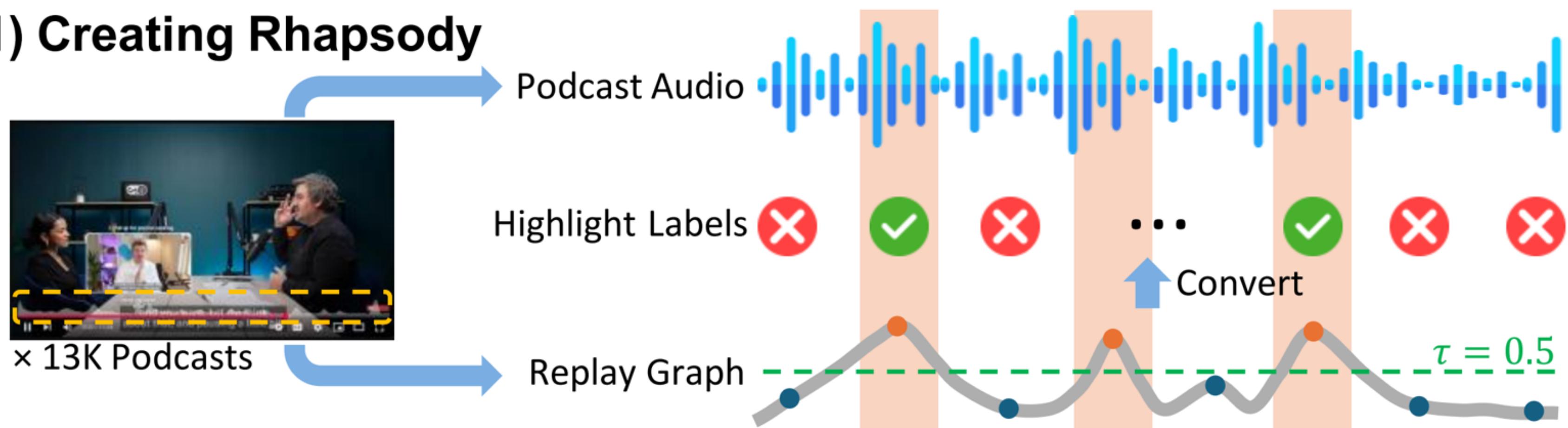


1 TL;DR

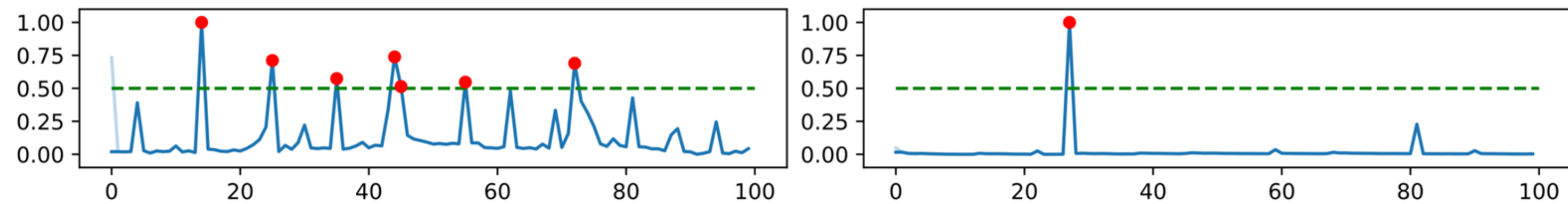
1. We introduce **RHAPSODY**, a dataset of 13K podcast episodes with segment-level highlight labels derived from YouTube's *most replayed* graphs.
2. We explore various systems to **predict highlights in long, unstructured spoken media**.

3 RHAPSODY Dataset

(1) Creating Rhapsody



- Contains **13,364** podcast episodes from YouTube.



- We use a peak-picking algorithm to **convert the raw replay graphs into segment-level, binary labels** indicating whether or not a segment is a highlight.

Task at a Glance

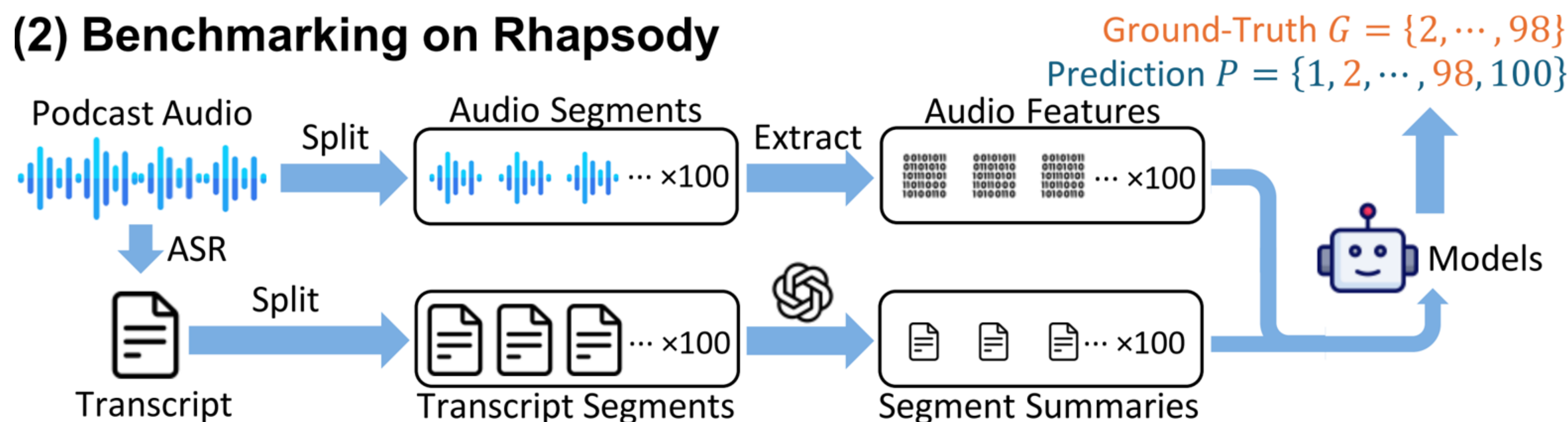
- **Input:** podcast audio A and 100 equal-length segments $\{c_i\}_{i=1}^{100}$, where c_i corresponds to $\frac{(i-1)d}{100}$ to $\frac{id}{100}$ second segment.
- **Output:** a set of highlight indices $P = \{p_k\}_{k=1}^{N_p}$
- **Metrics:** retrieval metrics, e.g., precision, recall, F1, ...

2 Motivation

- Podcasts are long (**avg. 30 min, 5K words**) and **unstructured** — hard to skim.
- Video highlight detection is well-studied — **why not for podcasts?**
- Usefulness: (1) save time (2) understand people's interest

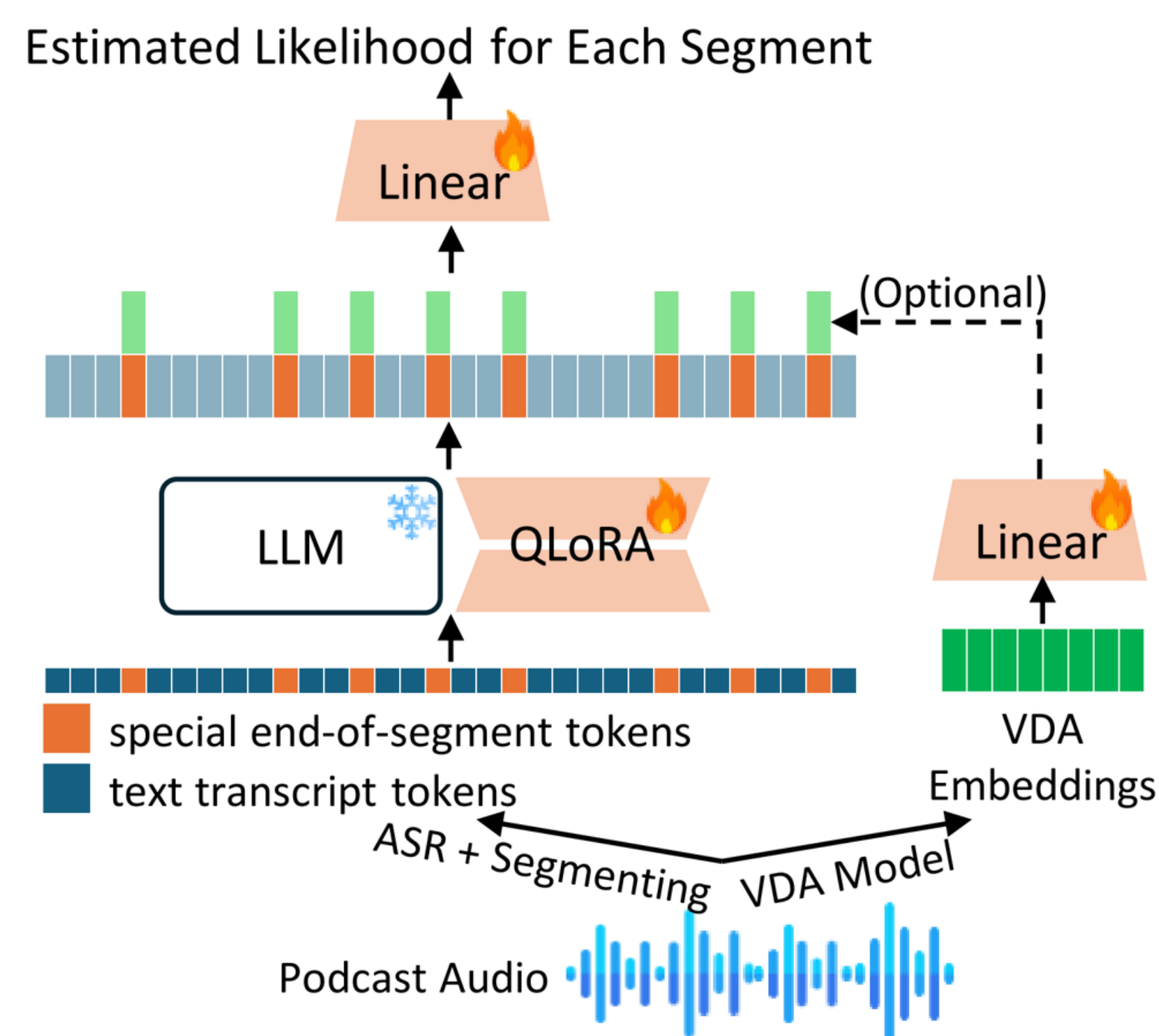
4 Input Data Preprocessing

(2) Benchmarking on Rhapsody



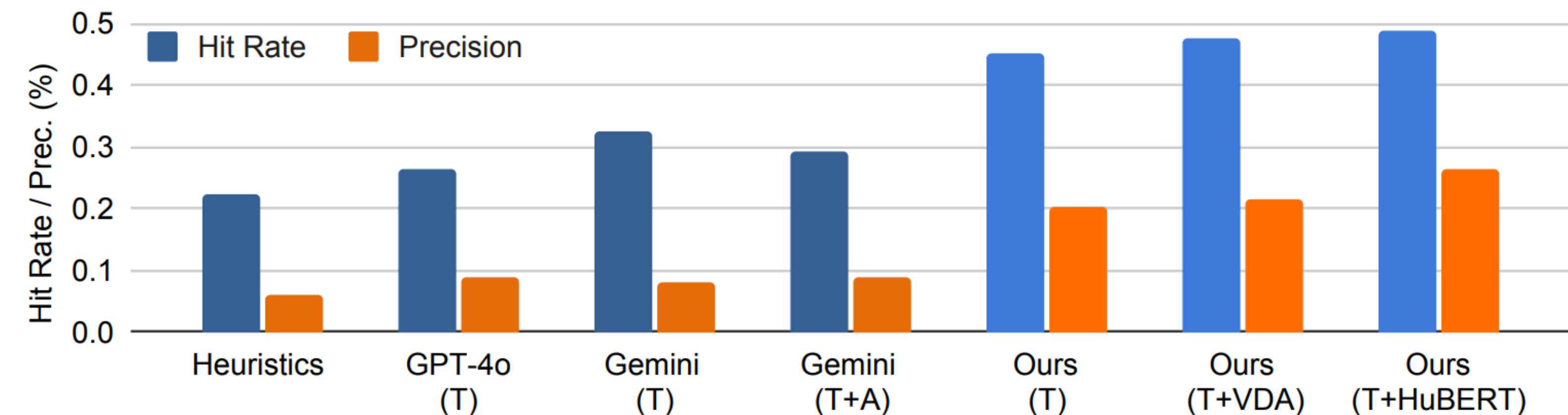
- **Audio Features:** Use **DVA** (Dominance, Valence, Arousal) or **HuBERT** embeddings

5 Fine-Tuning LMs

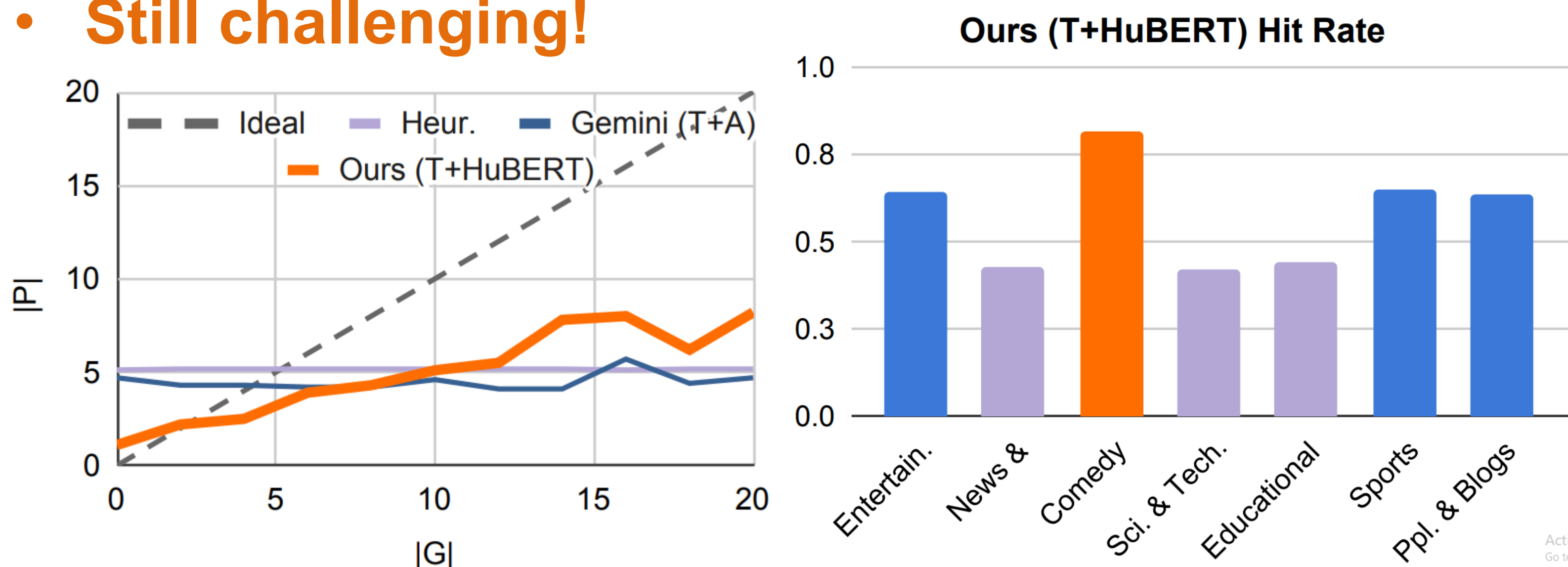


- **Llama 1B + QLoRa**
- Replace next-token prediction head with **segment-level binary classification head**
- **Audio features can be incorporated**
- **5.6M / 6.2M trainable params** in text-only / text + audio settings

6 Experimental Results



- **State-of-the-art LLMs struggle with this task.**
- **Fine-tuned Llama outperforms** other methods.
- Incorporating **audio features boosts performance.**
- **Still challenging!**



- # of predictions: **fine-tuned Llama adapts well**, but other systems fail
- There was **performance difference depending on podcast genres**.

7 Future Work

- Adaptive strategies depending on podcast genre
- Joint optimization in end-to-end framework
- Incorporating richer contextual cues
- Data bias & distribution analysis

